Unit 1

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Security goals

Confidentiality

Confidentiality is probably the most common aspect of information security. We need to protect our confidential information. An organization needs to guard against those malicious actions that endanger the confidentiality of its information. In the military, concealment of sensitive information is the major concern. In industry, hiding some information from competitors is crucial to the operation of the organization. In banking, customers' accounts need to be kept secret. As we will see later in this chapter, confidentiality not only applies to the storage of the information, it also applies to the transmission of information. When we send a piece of information to be stored in a remote computer or when we retrieve a piece of information from a remote computer, we need to conceal it during transmission.

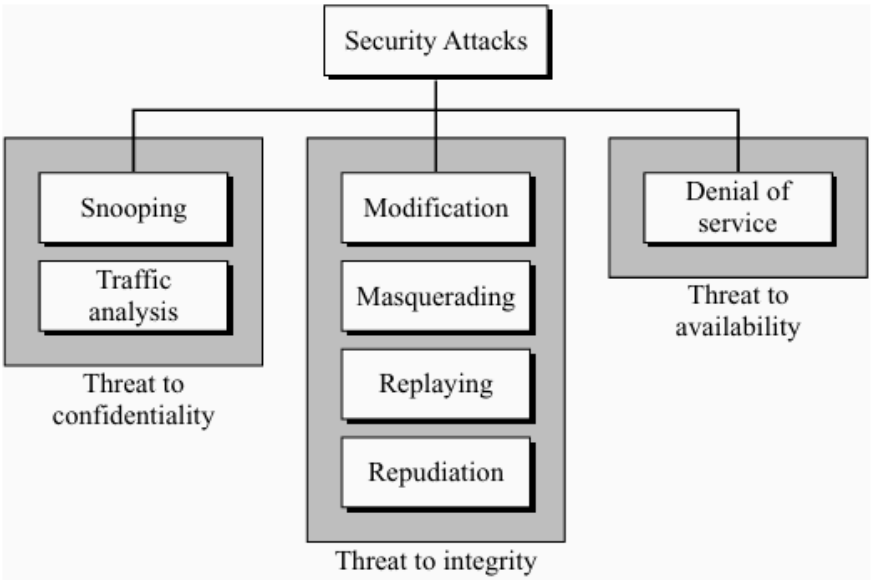
Integrity

Information needs to be changed constantly. In a bank, when a customer deposits or with draws money, the balance of her account needs to be changed. Integrity means that changes need to be done only by authorized entities and through authorized mechanisms. Integrity violation is not necessarily the result of a malicious act; an interruption in the system, such as a power surge, may also create unwanted changes in some information.

Availability

The third component of information security is availability. The information created and stored by an organization needs to be available to authorized entities. Information is use less if it is not available. Information needs to be constantly changed, which means it must be accessible to authorize entities. The unavailability of information is just as harmful for an organization as the lack of confidentiality or integrity. Imagine what would happen to a bank if the customers could not access their accounts for transactions.

Attacks



Attacks threatening confidentiality

Snooping

Snooping refers to unauthorized access to or interception of data. For example, a file transferred through the Internet may contain confidential information. An unauthorized entity may intercept the transmission and use the contents for her

own benefit. To prevent snooping, the data can be made nonintelligible to the interceptor by using encipherment techniques.

Traffic Analysis

Although encipherment of data may make it nonintelligible for the interceptor, she can obtain some other type information by monitoring online traffic. For example, she can find the electronic address (such as the e-mail address) of the sender or the receiver. She can collect pairs of requests and responses to help her guess the nature of transaction.

Attacks threatening integrity

Modification

After intercepting or accessing information, the attacker modifies the information to make it beneficial to herself. For example, a customer sends a message to a bank to do some transaction. The attacker intercepts the message and changes the type of transaction to benefit herself. Note that sometimes the attacker simply deletes or delays the message to harm the system or to benefit from it.

Masquerading

Masquerading, or spoofing, happens when the attacker impersonates somebody else. For example, an attacker might steal the bank card and PIN of a bank customer and pre tend that she is that customer. Sometimes the attacker pretends instead to be the receiver entity. For example, a user tries to contact a bank, but another site pretends that it is the bank and obtains some information from the user.

Replaying

Replaying is another attack. The attacker obtains a copy of a message sent by a user and later tries to replay it. For example, a person sends a request to her bank to ask for payment to the attacker, who has done a job for her. The attacker intercepts the message and sends it again to receive another payment from the bank.

Repudiation

This type of attack is different from others because it is performed by one of the two parties in the communication: the sender or the receiver. The sender of the message might later deny that she has sent the message; the receiver of the message might later deny that he has received the message. An example of denial by the sender would be a bank customer asking her bank to send some money to a third party but later denying that she has made such a request. An example of denial by the receiver could occur when a person buys a product from a manufacturer and pays for it electronically, but the manufacturer later denies having received the payment and asks to be paid.

Attacks threatening availability

Denial of service

Denial of service (DoS) is a very common attack. It may slow down or totally interrupt the service of a system. The attacker can use several strategies to achieve this. She might send so many bogus requests to a server that the server crashes because of the heavy load. The attacker might intercept and delete a server's response to a client, making the client to believe that the server is not responding. The attacker may also intercept requests from the clients, causing the clients to send requests many times and overload the system.

Passive versus active attacks

Attacks	*Passive/Active*	*Threatening*
Snooping Traffic analysis	Passive	Confidentiality
Modification Masquerading Replaying Repudiation	Active	Integrity
Denial of service	Active	Availability

Passive Attack

In a passive attack, the attacker's goal is just to obtain information. This means that the attack does not modify data or harm the system. The system continues with its normal operation. However, the attack may harm the sender or the receiver of the message. Attacks that threaten confidentiality-snooping and traffic analysis-are passive attacks. The revealing of the information may harm the sender or receiver of the message, but the system is not affected. For this reason, it is difficult to detect this type of attack until the sender or receiver finds out about the leaking of confidential information. Passive attacks, however, can be prevented by encipherment of the data.

Active Attack

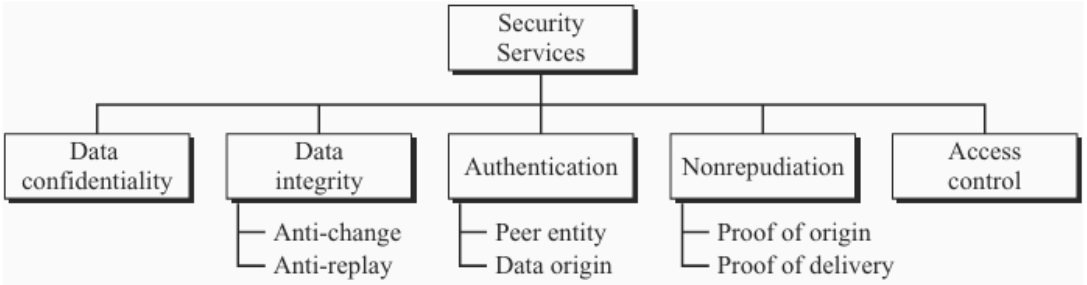
An active attack may change the data or harm the system. Attacks that threaten the integrity and availability are active attacks. Active attacks are normally easier to detect than to prevent, because an attacker can launch them in a variety of ways

Services and Mechanisms

The International Telecommunication Union-Telecommunication Standardization Sector (ITU-T) provides some security services and some mechanisms to implement those services.

Security Services

Figure 1.3 *Security services*



Data confidentiality

Data confidentiality is designed to protect data from disclosure attack. The service is very broad and encompasses confidentiality of the whole message or part of a message and also protection against traffic analysis. That is, it is designed to prevent snooping and traffic analysis attack.

Data integrity

Data integrity is designed to protect data from modification, insertion, deletion, and replaying by an adversary. It may protect the whole message or part of the message.

Authentication

This service provides the authentication of the party at the other end of the line. In connection oriented communication, it provides authentication of the sender or receiver during the connection establishment (peer entity authentication). In connectionless communication, it authenticates the source of the data (data origin authentication).

Nonrepudiation

Nonrepudiation service protects against repudiation by either the sender or the receiver of the data. In nonrepudiation with proof of the origin, the receiver of the data can later prove the identity of the sender if denied. In nonrepudiation

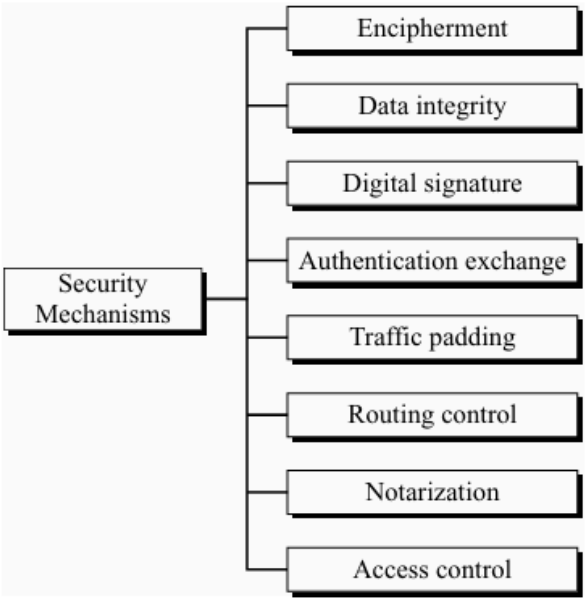
with proof of delivery, the sender of data can later prove that data were delivered to the intended recipient.

Access control

Access control provides protection against unauthorized access to data. The term access in this definition is very broad and can involve reading, writing, modifying, executing programs, and so on

Security mechanisms

Figure 1.4 *Security mechanisms*



Encipherment

Encipherment, hiding or covering data, can provide confidentiality. It can also be used to complement other mechanisms to provide other services. Today two techniques- cryptography and steganography- are used for enciphering.

Data integrity

The data integrity mechanism appends to the data a short checkvalue that has been created by a specific process from the data itself. The receiver receives the data and the checkvalue. He creates a new checkvalue from the received data and compares the newly created checkvalue with the one received. If the two checkvalues are the same, the integrity of data has been preserved.

Digital signature

A digital signature is a means by which the sender can electronically sign the data and the receiver can electronically verify the signature. The sender uses a process that involves showing that she owns a private key related to the public key that she has announced publicly. The receiver uses the sender's public key to prove that the message is indeed signed by the sender who claims to have sent the message.

Authentication exchange

In authentication exchange, two entities exchange some messages to prove their identity to each other. For example, one entity can prove that she knows a secret that only she is supposed to know.

Traffic padding

Traffic padding means inserting some bogus data into the data traffic to thwart the adversary's attempt to use the traffic analysis.

Routing control

Routing control means selecting and continuously changing different available routes between the sender and the receiver to prevent the opponent from eavesdropping on a particular route.

Notarization

Notarization means selecting a third trusted party to control the communication between two entities. This can be done, for example, to prevent repudiation. The receiver can involve a trusted party to store the sender request in order to prevent the sender from later denying that she has made such a request.

Access control

Access control uses methods to prove that a user has access right to the data or resources owned by a system. Examples of proofs are passwords and PINs.

Table 1.2 *Relation between security services and security mechanisms*

Security Service	*Security Mechanism*
Data confidentiality	Encipherment and routing control
Data integrity	Encipherment, digital signature, data integrity
Authentication	Encipherment, digital signature, authentication exchanges
Nonrepudiation	Digital signature, data integrity, and notarization
Access control	Access control mechanism

Techniques

Cryptography

Cryptography, a word with Greek origins, means “secret writing.” It is defined as involving three distinct mechanisms: symmetric-key encipherment, asymmetric-key encipherment, and hashing.

Symmetric-key encipherment

In symmetric-key encipherment (sometimes called secret-key encipherment or secret key cryptography), a single secret key for both encryption and decryption is used. Encryption/decryption can be thought of as electronic locking. In symmetric key enciphering, Alice puts the message in a box and locks the box using the shared secret key; Bob unlocks the box with the same key and takes out the message.

Asymmetric-key encipherment

In asymmetric-key encipherment (sometimes called public-key encipherment or public-key cryptography), we have the same situation as the symmetric-key encipherment, with a few exceptions. First, there are two keys instead of one: one public key and one private key. To send a secured message to Bob, Alice first encrypts the message using Bob's public key. To decrypt the message, Bob uses his own private key.

Hashing

In hashing, a fixed-length message digest is created out of a variable-length message. The digest is normally much smaller than the message. To be useful, both the message and the digest must be sent to Bob. Hashing is used to provide checkvalues.

Steganography

The word steganography, with origin in Greek, means "covered writing," in contrast with cryptography, which means "secret writing." Cryptography means concealing the contents of a message by enciphering; steganography means concealing the message itself by covering it with something else.

Text cover

The cover of secret data can be text. There are several ways to insert binary data into an innocuous text. For example, we can use single space between words to represent the binary digit 0 and double space to represent binary digit 1. The following short message hides the 8-bit binary representation of the letter A in ASCII code (01000001).

Image cover

Secret data can also be covered under a color image. Digitized images are made of pixels (picture elements), in which normally each pixel uses 24 bits (three bytes). Each byte represents one of the primary colors (red, green, or blue). We can there fore have 2^8 different shades of each color. In a method called LSB (least significant bit), the least significant bit of each byte is set to zero. This may make the image a little bit lighter in some areas, but this is not normally noticed. Now we can hide a binary data in the image by keeping or

changing the least significant bit. If our binary digit is 0, we keep the bit; if it is 1, we change the bit to 1. In this way, we can hide a character (eight ASCII bits) in three pixels.

Other covers

Other Covers Other covers are also possible. The secret message, for example, can be covered under audio (sound and music) and video. Both audio and video are compressed today; the secret data can be embedded during or before the compression.

Mathematics of cryptography

Integer arithmetic

Integer division

Two Restrictions
When we use the a = q*n + r division relationship in cryptography, we impose two restrictions. First, we require that the divisor be a positive integer (n > 0). Second, we require that the remainder be a nonnegative integer (r ≥ 0).

Example 2.3

When we use a computer or a calculator, r and q are negative when a is negative. How can we apply the restriction that r needs to be positive? The solution is simple, we decrement the value of q by 1 and we add the value of n to r to make it positive.

$$-255 = (-\mathbf{23} \times 11) + (-\mathbf{2}) \qquad \leftrightarrow \qquad -255 = (-\mathbf{24} \times 11) + \mathbf{9}$$

We have decremented –23 to become –24 and added 11 to –2 to make it 9. The above relation is still valid.

Divisibility

Example 2.4

a. The integer 4 divides the integer 32 because $\mathbf{32} = 8 \times \mathbf{4}$. We show this as $4|32$.
b. The number 8 does not divide the number 42 because $\mathbf{42} = 5 \times \mathbf{8} + 2$. There is a remainder, the number 2, in the equation. We show this as $8 \nmid 42$.

Example 2.5

a. We have $13|78$, $7|98$, $-6|24$, $4|44$, and $11|(-33)$.
b. We have $13 \nmid 27$, $7 \nmid 50$, $-6 \nmid 23$, $4 \nmid 41$, and $11 \nmid (-32)$.

Properties

Following are several properties of divisibility. The interested reader can check Appendix Q for proofs.

Property 1: if $a|1$, then $a = \pm 1$.
Property 2: if $a|b$ and $b|a$, then $a = \pm b$.
Property 3: if $a|b$ and $b|c$, then $a|c$.
Property 4: if $a|b$ and $a|c$, then $a|(m \times b + n \times c)$, where m and n are arbitrary integers.

Example 2.6

a. Since $3|15$ and $15|45$, according to the third property, $3|45$.
b. Since $3|15$ and $3|9$, according to the fourth property, $3|(15 \times 2 + 9 \times 4)$, which means $3|66$.

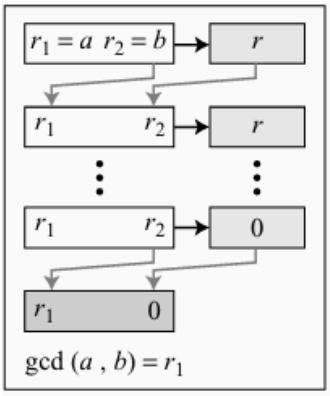
Greatest common divisor

The greatest common divisor of two positive integers is the largest integer that can divide both integers.

Euclidean algorithm

Fact 1: $\gcd(a, 0) = a$
Fact 2: $\gcd(a, b) = \gcd(b, r)$, where r is the remainder of dividing a by b

Figure 2.7 *Euclidean algorithm*



a. Process

```
r1 ← a; r2 ← b;   (Initialization)
while (r2 > 0)
  {
   q ← r1 / r2;
   r ← r1 − q × r2;
   r1 ← r2; r2 ← r;
  }

   gcd (a, b) ← r1
```

b. Algorithm

When gcd (a, b) = 1, we say that a and b are relatively prime.

Example 2.7

Find the greatest common divisor of 2740 and 1760.

Solution

We apply the above procedure using a table. We initialize r_1 to 2740 and r_2 to 1760. We have also shown the value of q in each step. We have gcd (2740, 1760) = 20.

q	r_1	r_2	r
1	2740	1760	980
1	1760	980	780
1	980	780	200
3	780	200	180
1	200	180	20
9	180	20	0
	20	0	

The Extended Euclidean Algorithm

Given two integers a and b, we often need to find other two integers, s and t, such that

$$s \times a + t \times b = \gcd(a, b)$$

```
 r1 ← a; r2 ← b;
 s1 ← 1; s2 ← 0;   (Initialization)
 t1 ← 0; t2 ← 1;
while (r2 > 0)
 {
  q ← r1 / r2;
   r  ← r1 − q × r2;
   r1 ← r2;  r2 ← r;   (Updating r's)
   s  ← s1 − q × s2;
   s1 ← s2;  s2 ← s;   (Updating s's)
   t  ← t1 − q × t2;
   t1 ← t2;  t2 ← t;   (Updating t's)
 }
  gcd (a , b) ← r1;   s ← s1;   t ← t1
```

b. Algorithm

Example 2.9

Given $a = 161$ and $b = 28$, find gcd (a, b) and the values of s and t.

Solution

$$r = r_1 - q \times r_2 \qquad s = s_1 - q \times s_2 \qquad t = t_1 - q \times t_2$$

We use a table to follow the algorithm.

q	r_1	r_2	r	s_1	s_2	s	t_1	t_2	t
5	161	28	21	1	0	1	0	1	−5
1	28	21	7	0	1	−1	1	−5	6
3	21	7	0	1	−1	4	−5	6	−23
	7	0		**−1**	4		**6**	−23	

We get gcd (161, 28) = 7, $s = -1$ and $t = 6$. The answers can be tested because we have

$$(-1) \times 161 + 6 \times 28 = 7$$

Linear Diophantine equations

Although we will see a very important application of the extended Euclidean algorithm in the next section, one immediate application is to find the solutions to the linear Diophantine equations of two variables, an equation of type ax + by = c. We need to find integer values for x and y that satisfy the equation. This type of equation has either no solution or an infinite number of solutions.
Let d = gcd (a, b). If d !| c, then the equation has no solution. If d| c, then we have an infinite number of solutions. One of them is called the particular; the rest, general.

Particular Solution

If d| c, a particular solution to the above equation can be found using the following steps:

1. Reduce the equation to $a_1x + b_1y = c_1$ by dividing both sides of the equation by d. This is possible because d divides a, b, and c by the assumption.
2. Solve for s and t in the relation $a_1s + b_1t = 1$ using the extended Euclidean algorithm.
3. The particular solution can be found:

Particular solution: $x_0 = (c/d)s$ ***and*** $y_0 = (c/d)t$

General Solutions

After finding the particular solution, the general solutions can be found:

General solutions: $x = x_0 + k\,(b/d)$ ***and*** $y = y_0 - k\,(a/d)$ **where *k* is an integer**

Example 2.12

Find the particular and general solutions to the equation $21x + 14y = 35$.

Solution

We have d = gcd (21, 14) = 7. Since 7|35, the equation has an infinite number of solutions. We can divide both sides by 7 to find the equation $3x + 2y = 5$. Using the extended Euclidean algorithm, we find s and t such as $3s + 2t = 1$. We have $s = 1$ and $t = -1$. The solutions are

Particular: $x_0 = 5 \times 1 = 5$ and $y_0 = 5 \times (-1) = -5$	since 35/7 = 5
General: $x = 5 + k \times 2$ and $y = -5 - k \times 3$	where k is an integer

Therefore, the solutions are (5, −5), (7, −8), (9, −11), . . . We can easily test that each of these solutions satisfies the original equation.

Modular Arithmetic

Modulo operator

The modulo operator (mod) takes an integer (a) from the set Z and a positive modulus (n). The operator creates a nonnegative residue (r).

Example 2.14

Find the result of the following operations:

a. 27 mod 5
b. 36 mod 12
c. −18 mod 14
d. −7 mod 10

Solution

We are looking for the residue r. We can divide the a by n and find q and r. We can then disregard q and keep r.

a. Dividing 27 by 5 results in $r = 2$. This means that 27 mod 5 = 2.
b. Dividing 36 by 12 results in $r = 0$. This means that 36 mod 12 = 0.
c. Dividing −18 by 14 results in $r = -4$. However, we need to add the modulus (14) to make it nonnegative. We have $r = -4 + 14 = 10$. This means that −18 mod 14 = 10.
d. Dividing −7 by 10 results in $r = -7$. After adding the modulus to −7, we have $r = 3$. This means that −7 mod 10 = 3.

Congruence

To show wthat two integers are congruent, we use the congruence operator (≡). We add the
phrase (mod n) to the right side of the congruence to define the value of modulus that makes the relationship valid.

$2 \equiv 12 \pmod{10}$	$13 \equiv 23 \pmod{10}$	$34 \equiv 24 \pmod{10}$	$-8 \equiv 12 \pmod{10}$
$3 \equiv 8 \pmod{5}$	$8 \equiv 13 \pmod{5}$	$23 \equiv 33 \pmod{5}$	$-8 \equiv 2 \pmod{5}$

The congruence operator looks like the equality operator, but there are differences. First, an equality operator maps a member of Z to itself; the

congruence operator maps a member from Z to a member of Zn. Second, the equality operator is one to-one; the congruence operator is many-to one.

The phrase (mod n) that we insert at the right-hand side of the congruence operator is just an indication of the destination set (Zn). We need to add this phrase to show what modulus is used in the mapping. The symbol mod used here does not have the same meaning as the binary operator. In other words, the symbol mod in 12 mod 10 is an operator; the phrase (mod 10) in 2 ≡ 12 (mod 10) means that the destination set is Z10.

Residue classes

A residue class [a] or [a]n is the set of integers congruent modulo n. In other words, it is the set of all integers such that x = a (mod n). For example, if n = 5, we have five sets [0], [1], [2], [3], and [4] as shown below:

The set Zn is the set of all least residue modulo n.

Operations in Zn

Properties

First Property:	$(a + b) \bmod n = [(a \bmod n) + (b \bmod n)] \bmod n$
Second Property:	$(a - b) \bmod n = [(a \bmod n) - (b \bmod n)] \bmod n$
Third Property:	$(a \times b) \bmod n = [(a \bmod n) \times (b \bmod n)] \bmod n$

Example 2.19

In arithmetic, we often need to find the remainder of powers of 10 when divided by an integer. For example, we need to find 10 mod 3, 10^2 mod 3, 10^3 mod 3, and so on. We also need to find 10 mod 7, 10^2 mod 7, 10^3 mod 7, and so. The third property of the mod operator mentioned above makes life much easier.

$10^n \bmod x = (10 \bmod x)^n \bmod x$ Applying the third property n times.

We have

$$10 \bmod 3 = 1 \rightarrow 10^n \bmod 3 = (10 \bmod 3)^n = 1$$
$$10 \bmod 9 = 1 \rightarrow 10^n \bmod 9 = (10 \bmod 9)^n = 1$$
$$10 \bmod 7 = 3 \rightarrow 10^n \bmod 7 = (10 \bmod 7)^n = 3^n \bmod 7$$

Inverses

In modular arithmetic, each integer has an additive inverse. The sum of an integer and its additive inverse is congruent to 0 modulo n.

In modular arithmetic, an integer may or may not have a multiplicative inverse. When it does, the product of the integer and its multiplicative inverse is congruent to 1 modulo n.

It can be proved that a has a multiplicative inverse in Zn if and only if gcd(n,a) = 1
In this case, a and n are said to be relatively prime.

$$(s \times n + b \times t) \bmod n = 1 \bmod n$$
$$[(s \times n) \bmod n] + [(b \times t) \bmod n] = 1 \bmod n$$
$$0 + [(b \times t) \bmod n] = 1$$
$$(b \times t) \bmod n = 1 \qquad \rightarrow \text{This means } t \text{ is the multiplicative inverse of } b \text{ in } Z_n$$

The extended Euclidean algorithm finds the multiplicative inverses of b in Zn when n and b are given and gcd (n, b) = 1.
The multiplicative inverse of b is the value of t after being mapped to Zn.

Example 2.26

Find the multiplicative inverse of 23 in $\mathbf{Z_{100}}$.

Solution

We use a table similar to the one we used before with $r_1 = 100$ and $r_2 = 23$. We are interested only in the value of t.

q	r_1	r_2	r	t_1	t_2	t
4	100	23	8	0	1	−4
2	23	8	7	1	−4	19
1	8	7	1	−4	9	−13
7	7	1	0	9	−13	100
	1	0		−13	100	

The gcd (100, 23) is 1, which means the inverse of 23 exists. The extended Euclidean algorithm gives $t_1 = -13$. The inverse is $(-13) \bmod 100 = 87$. In other words, 13 and 87 are multiplicative inverses in $\mathbf{Z_{100}}$. We can see that $(23 \times 87) \bmod 100 = 2001 \bmod 100 = 1$.

Figure 2.16 *Addition and multiplication tables for $\mathbf{Z}_{10}$*

	0	1	2	3	4	5	6	7	8	9
0	0	1	2	3	4	5	6	7	8	9
1	1	2	3	4	5	6	7	8	9	0
2	2	3	4	5	6	7	8	9	0	1
3	3	4	5	6	7	8	9	0	1	2
4	4	5	6	7	8	9	0	1	2	3
5	5	6	7	8	9	0	1	2	3	4
6	6	7	8	9	0	1	2	3	4	5
7	7	8	9	0	1	2	3	4	5	6
8	8	9	0	1	2	3	4	5	6	7
9	9	0	1	2	3	4	5	6	7	8

Addition Table in $\mathbf{Z}_{10}$

	0	1	2	3	4	5	6	7	8	9
0	0	0	0	0	0	0	0	0	0	0
1	0	1	2	3	4	5	6	7	8	9
2	0	2	4	6	8	0	2	4	6	8
3	0	3	6	9	2	5	8	1	4	7
4	0	4	8	2	6	0	4	8	2	6
5	0	5	0	5	0	5	0	5	0	5
6	0	6	2	8	4	0	6	2	8	4
7	0	7	4	1	8	0	2	9	6	3
8	0	8	6	4	2	0	8	6	4	2
9	0	9	8	7	6	5	4	3	2	1

Multiplication Table in $\mathbf{Z}_{10}$

inverse. We need another set. The new set, which is a subset of $\mathbf{Z}_n$ includes only integers in $\mathbf{Z}_n$ that have a unique multiplicative inverse. This set is called $\mathbf{Z}_{n^*}$. Figure 2.17 shows some instances of two sets. Note that $\mathbf{Z}_{n^*}$ can be made from multiplication tables such as the one shown in Figure 2.16.

Each member of $\mathbf{Z}_n$ has an additive inverse, but only some members have a multiplicative inverse. Each member of $\mathbf{Z}_{n^*}$ has a multiplicative inverse, but only some members have an additive inverse.

We need to use $\mathbf{Z}_n$ when additive inverses are needed; we need to use $\mathbf{Z}_{n^*}$ when multiplicative inverses are needed.

Two More Sets

Cryptography often uses two more sets: $\mathbf{Z}_p$ and $\mathbf{Z}_{p^*}$. The modulus in these two sets is a prime number. Prime numbers will be discussed in later chapters; suffice it to say that a prime number has only two divisors: integer 1 and itself.

The set $\mathbf{Z}_p$ is the same as $\mathbf{Z}_n$ except that n is a prime. $\mathbf{Z}_p$ contains all integers from 0 to $p - 1$. Each member in $\mathbf{Z}_p$ has an additive inverse; each member except 0 has a multiplicative inverse.

The set $\mathbf{Z}_{p^*}$ is the same as $\mathbf{Z}_{n^*}$ except that n is a prime. $\mathbf{Z}_{p^*}$ contains all integers from 1 to $p - 1$. Each member in $\mathbf{Z}_{p^*}$ has an additive and a multiplicative inverse. $\mathbf{Z}_{p^*}$ is a very good candidate when we need a set that supports both additive and multiplicative inverse.

The following shows these two sets when $p = 13$.

$Z_{13} = \{0, 1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12\}$
$Z_{13}* = \{1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12\}$

Matrices

An additive identity matrix, denoted as 0, is a matrix with all rows and columns set to 0's. An identity matrix, denoted as I, is a square matrix with 1s on the main diagonal and 0s elsewhere.

Residue matrices

Cryptography uses residue matrices: matrices with all elements are in Zn. All operations on residue matrices are performed the same as for the integer matrices except that the operations are done in modular arithmetic. One interesting result is that a residue matrix has a multiplicative inverse if the determinant of the matrix has a multiplicative inverse in Zn In other words, a residue matrix has a multiplicative inverse if gcd (det(A), n) = 1.

Congruence

Two matrices are congruent modulo n, written as $\mathbf{A} \equiv \mathbf{B} \pmod{n}$, if they have the same number of rows and columns and all corresponding elements are congruent modulo n. In other words, $\mathbf{A} \equiv \mathbf{B} \pmod{n}$ if $a_{ij} \equiv b_{ij} \pmod{n}$ for all i's and j's.

Linear congruence

Cryptography often involves solving an equation or a set of equations of one or more variables with coefficient in $\mathbf{Z}_n$. This section shows how to solve equations when the power of each variable is 1 (linear equation).

Single-Variable Linear Equations

Let us see how we can solve equations involving a single variable—that is, equations of the form $ax \equiv b \pmod{n}$. An equation of this type might have no solution or a limited number of solutions. Assume that the gcd $(a, n) = d$. If $d \nmid b$, there is no solution. If $d|b$, there are d solutions.

If $d|b$, we use the following strategy to find the solutions:

1. Reduce the equation by dividing both sides of the equation (including the modulus) by d.
2. Multiply both sides of the reduced equation by the multiplicative inverse of a to find the particular solution x_0.
3. The general solutions are $x = x_0 + k\,(n/d)$ for $k = 0, 1, \ldots, (d - 1)$.

Example 2.36

Solve the equation $14x \equiv 12 \pmod{18}$.

Solution

Note that gcd (14 and 18) = 2. Since 2 divides 12, we have exactly two solutions, but first we reduce the equation.

$$14x \equiv 12 \pmod{18} \rightarrow \quad 7x \equiv 6 \pmod{9} \quad \rightarrow x \equiv 6\,(7^{-1}) \pmod{9}$$
$$x_0 = (6 \times 7^{-1}) \bmod 9 = (6 \times 4) \pmod{9} = 6$$
$$x_1 = x_0 + 1 \times (18/2) = 15$$

Both solutions, 6 and 15 satisfy the congruence relation, because $(14 \times 6) \bmod 18 = 12$ and also $(14 \times 15) \bmod 18 = 12$.